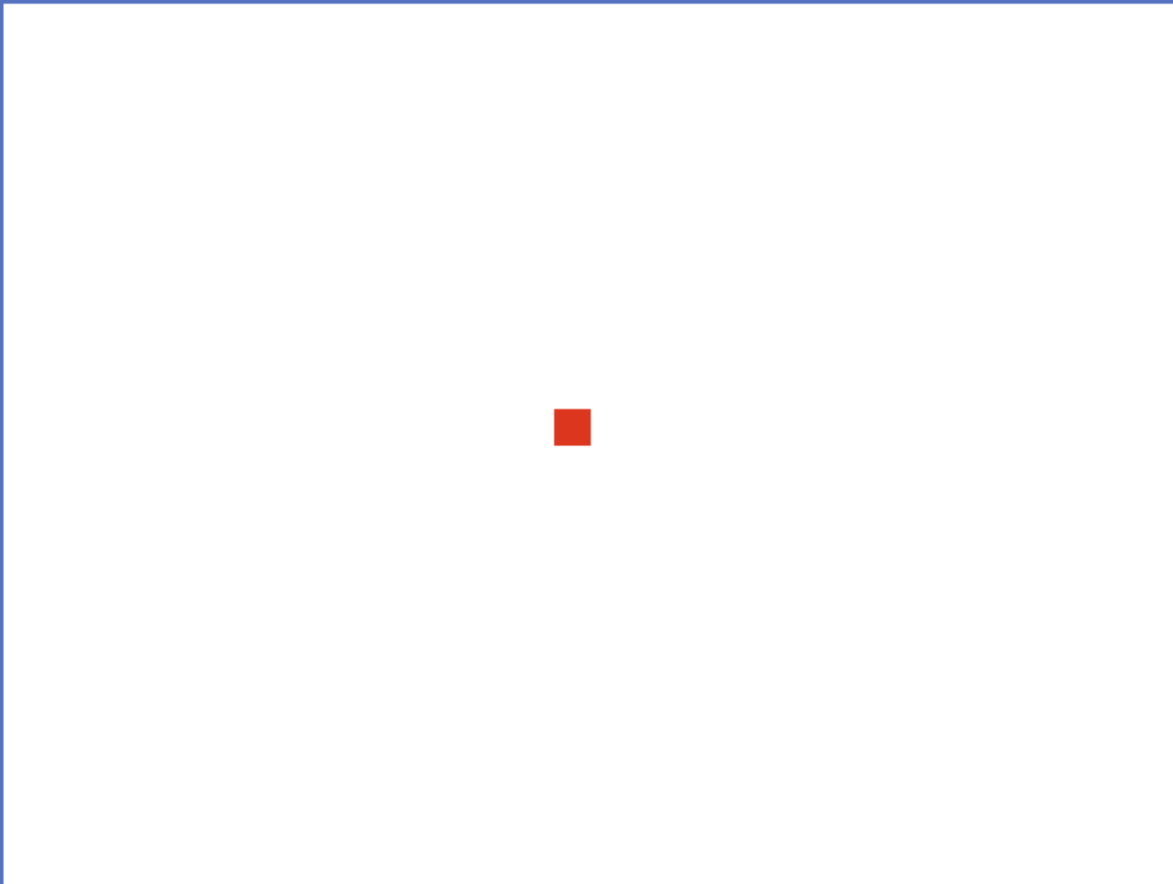


10.1

```
1|      mov R2, #1          ; 1sec delay
2|flash:
3|      MOV R0, #.red
4|      MOV R1, R2
5|      BL drawpixel        ; on
6|      MOV R0, #.white
7|      MOV R1, R2
8|      BL drawpixel        ; off
9|      B flash
10|     HALT
11|;-----
12|drawpixel:
13|     PUSH {R3,R4}
14|     MOV R3, R0            ; copy pixel colour to R3
15|     MOV R4, R1            ; copy delay time to R4
16|     STR R3, .Pixel367    ; draw pixel with colour
17|     PUSH {R0, LR}
18|     MOV R0, R4            ; pass delay time to delay function
19|     BL delay              ; call delay function
20|     POP {R0, LR}         ; restore R0 and LR after function call
21|     POP {R3,R4}
22|     RET
23|;-----
24|delay:
25|     push {R3,R4,R5,R6}
26|     MOV R3, R0            ; move delay into R3
27|     LDR R4, .Time        ; get start ime
28|timer:
29|     LDR R5, .Time        ; update time
30|     SUB R6, R5, R4        ; calculate elapsed time
31|     CMP R6, R3           ; compare elapsed to delay time
32|     BLT timer
33|     pop {R3,R4,R5,R6}
34|     RET
```

Input/Output



c) PUSH {LR}, because when calling a BL after another BL, the current LR will be overwritten with the new LR, so if you want to save the old LR, you have to use PUSH {LR}

Program

```

1 |      MOV R2, #1          ; 1 sec delay
2 |      MOV R7, #1          ; only 1 sec pause because 1 plus original 1 sec delay
3 |      MOV R8, #3          ; 3 times counter
4 |      MOV R9, #0          ; placeholder
5 | flash:
6 |      MOV R0, #.red
7 |      MOV R1, R2
8 |      BL drawpixel        ; on for 1 sec
9 |      MOV R0, #.white
10 |     MOV R1, R2
11 |     BL drawpixel        ; off for 1 sec
12 |     ADD R9, R9, #1       ; increment by 1
13 |     CMP R9, R8           ; compare the current increment to 3
14 |     BNE flash           ; repeat 3 times
15 |     MOV R0, R7
16 |     BL delay            ; 2 sec pause
17 |     MOV R9, #0
18 |     B flash
19 |     HALT
20 | ;-----
21 | drawpixel:
22 |     PUSH {R3,R4}
23 |     MOV R3, R0
24 |     MOV R4, R1
25 |     STR R3, .Pixel367
26 |     PUSH {R0, LR}
27 |     MOV R0, R4
28 |     BL delay
29 |     POP {R0, LR}
30 |     POP {R3,R4}
31 |     RET
32 | ;-----
33 | delay:
34 |     PUSH {R3,R4,R5,R6}
35 |     MOV R3, R0
36 |     LDR R4, .Time

```

10.3

```
1|      MOV R0, #3          ; number of flashes
2|      MOV R1, #2          ; pause time
3|      BL flashpattern
4|      HALT
5| ;-----
6| flashpattern:
7|      PUSH {R2, R7, R8, R9, LR}
8|      MOV R8, R0          ; move flash count
9|      MOV R7, R1          ; move pause time
10|     MOV R2, #1           ; 1 second delay
11|     MOV R9, #0           ; counter
12| flash:
13|     MOV R0, #.red
14|     MOV R1, R2
15|     BL drawpixel        ; on for 1 sec
16|     MOV R0, #.white
17|     MOV R1, R2
18|     BL drawpixel        ; off for 1 sec
19|     ADD R9, R9, #1       ; increment by 1
20|     CMP R9, R8           ; compare the current increment to 3
21|     BNE flash           ; repeat 3 times
22|     MOV R0, R7
23|     BL delay            ; 2 sec pause
24|     MOV R9, #0          ; reset counter
25|     B flash
26|     HALT
27| ;-----
28| drawpixel:
29|     PUSH {R3,R4}
30|     MOV R3, R0
31|     MOV R4, R1
32|     STR R3, .Pixel367
33|     PUSH {R0, LR}
34|     MOV R0, R4
35|     BL delay
36|     POP {R0, LR}
37|     POP {R3,R4}
```
